KGiSLInstituteofTechnology

(AnAutonomous Institution)

Affiliated to Anna University, Approved by AICTE,
Recognized by UGC, Accredited by **NAAC** & **NBA** (B.E-CSE,B.E-ECE, B.Tech-IT),365, KGiSL Campus, Thudiyalur Road,
Saravanampatti,Coimbatore – 641035.

DEPARTMENTOFCOMPUTERSCIENCE AND ENGINEERING

CCS356 - OBJECT ORIENTED SOFTWARE ENGINEERING

ASSIGNMENT - 2

TOPIC:

Railway Reservation System

Submitted By

NAME : ASHWIN S

ROLL NO : 23CSA14

REG NO :711723104014

Railway Reservation System

1. Introduction

1.1 Purpose

The purpose of the Order Management System (OMS) is to provide an automated platform for managing customer orders, product inventory, payment processing, and delivery tracking. The system helps businesses efficiently handle order-related operations and improve customer service.

1.2 Scope

The Order Management System allows customers to browse products, place orders, track order status, and make payments online. It also enables administrators to manage products, update inventory, process orders, and generate reports.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
OMS	Order Management System
Admin	System administrator who manages products and orders
Customer	User who purchases products
Order ID	Unique identifier for each order
Inventory	Stock of products available for sale

1.4 Document Overview

This document describes the overall system design, functional and non-functional requirements, system architecture, database design, and security features of the Order Management System.

2. Overall Description

The Order Management System is a web-based application designed to manage product orders in an efficient and organized manner. Customers can browse products, place orders, and track deliveries. Administrators can manage products, update stock levels, and monitor order processing. The system ensures fast order processing, accurate inventory tracking, and secure payment handling.

3. System Features

The main features of the Order Management System include:

- User registration and login
- Product browsing and searching
- Online order placement
- Secure payment processing
- Order tracking and history
- Inventory management
- Administrative control panel
- Report generation for sales and orders

4. Functional Requirements

The system should:

- Allow customers to register and log in securely
- Display available products with price and description
- Enable customers to add products to the shopping cart
- Allow customers to place orders and generate Order IDs
- Support online payment through payment gateways
- Allow customers to view order status and history
- Enable administrators to add, update, or remove products
- Manage product inventory automatically

- Generate sales reports for administrators

5. Non-Functional Requirements

The system must ensure:

- High performance with quick response time
- Secure authentication and authorization
- Data accuracy and reliability
- User-friendly interface
- 24/7 availability
- Backup and recovery mechanisms
- Scalability to support large numbers of users

6. External Interface Requirements

User Interface

The system provides web-based interfaces for customers and administrators.

Hardware Interface

Users can access the system through computers, tablets, or smartphones connected to the internet.

Software Interface

The system interacts with:

- Database management system
- Payment gateway
- Web browser

Communication Interface

Communication occurs through secure internet protocols such as HTTPS.

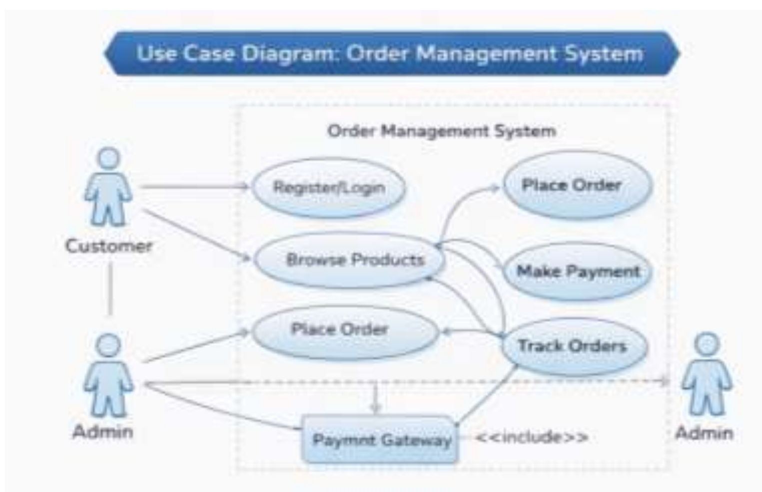
7. System Architecture

The system follows a **three-tier architecture**:

1. **Presentation Layer**
    - User interface for customers and administrators
2. **Application Layer**
    - Business logic for order processing, payment, and inventory management
3. **Data Layer**
    - Database storing users, products, orders, and payments

8. UML Diagrams

Use Case Diagram



Actors:

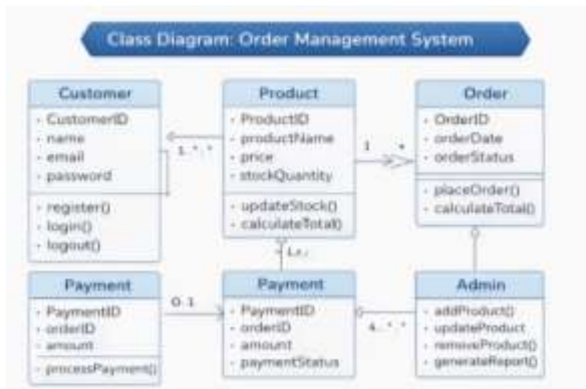
- Customer
- Admin
- Payment Gateway

Use Cases:

- Register/Login

- Browse Products
- Place Order
- Make Payment
- Track Order
- Manage Products (Admin)

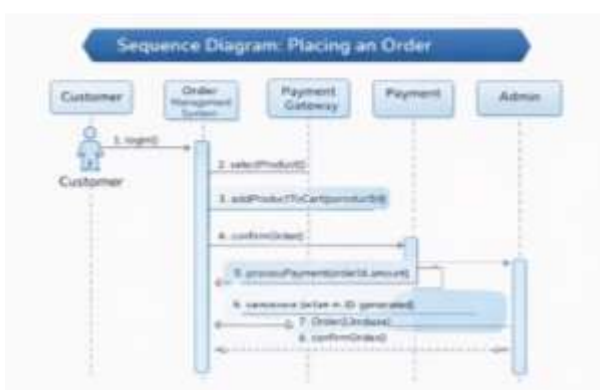
Class Diagram



Main Classes:

- Customer
- Product
- Order
- Payment
- Admin

Sequence Diagram



Steps for placing an order:

1. Customer logs in
2. Selects product
3. Adds product to cart
4. Confirms order
5. Payment gateway processes payment
6. System generates Order ID
7. Order confirmation is displayed

9. Database Design

Main tables include:

Customer Table

- CustomerID
- Name
- Email
- Password
- Phone

Product Table

- ProductID
- ProductName

- Price
- StockQuantity

Order Table

- OrderID
- CustomerID
- OrderDate
- OrderStatus

Payment Table

- PaymentID
- OrderID
- Amount
- PaymentStatus

10. Security Requirements

The system ensures security through:

- User authentication and authorization
- Password encryption
- Secure payment gateway integration
- Data protection and privacy
- Regular system monitoring

11. Future Enhancements

Possible improvements include:

- Mobile application support
- AI-based product recommendations
- Real-time delivery tracking

- Integration with external logistics systems
- Advanced analytics and reporting

12. Conclusion

The Order Management System provides a reliable and efficient platform for managing customer orders, inventory, and payments. By automating order processing and improving inventory management, the system enhances business productivity and customer satisfaction.